

Database Systems Lab

Lab Task 5



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24P-0554

Section 4C

Operation 01

```
.mode table
.system cls

SELECT c.first_name || ' ' || c.last_name AS "Customer Full Name",
       o.order_date AS "Order Date"
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id;
```

```
+-----+-----+
| Customer Full Name | Order Date |
+-----+-----+
| Ali Khan           | 2023-10-01 |
| Sara Ahmed         | 2023-10-05 |
| Ali Khan           | 2023-11-10 |
| John Smith         | 2023-11-15 |
| Maria Garcia       | 2023-12-01 |
+-----+-----+
sqlite> █
```

Operation 02

```
SELECT p.product_name AS "Product Name",
       cat.category_name AS "Category Name"
FROM products p
JOIN categories cat ON p.category_id = cat.category_id;
```

```
+-----+-----+
| Product Name       | Category Name |
+-----+-----+
| Laptop Pro 15      | Electronics   |
| Wireless Mouse     | Electronics   |
| Database Design Book | Books        |
| Winter Jacket      | Clothing      |
| Microwave Oven     | Home Appliances |
| Smartphone X       | Electronics   |
| Novel: The Last Query | Books        |
+-----+-----+
sqlite> █
```

Operation 03

```
SELECT oi.order_id AS "Order ID",
       p.product_name AS "Product Name",
       oi.quantity AS "Quantity"
FROM order_items oi
JOIN products p ON oi.product_id = p.product_id;
```

Order ID	Product Name	Quantity
1	Laptop Pro 15	1
1	Wireless Mouse	2
2	Database Design Book	1
3	Smartphone X	1
3	Wireless Mouse	1
4	Novel: The Last Query	3
5	Winter Jacket	1

Operation 04

```
SELECT p.payment_id AS "Payment ID",
       p.amount AS "Amount",
       p.payment_date AS "Payment Date",
       p.payment_method AS "Payment Method",
       c.first_name || ' ' || c.last_name AS "Customer Name"
FROM payments p
JOIN orders o ON p.order_id = o.order_id
JOIN customers c ON o.customer_id = c.customer_id;
```

Payment ID	Amount	Payment Date	Payment Method	Customer Name
1	1250.0	2023-10-01	Credit Card	Ali Khan
2	925.0	2023-11-10	PayPal	Ali Khan
3	150.0	2023-12-01	Credit Card	Maria Garcia

Operation 05

```
SELECT c.customer_id,  
       c.first_name || ' ' || c.last_name AS "Customer Name",  
       SUM(oi.quantity * oi.unit_price) AS "Total Amount Spent"  
FROM customers c  
LEFT JOIN orders o ON c.customer_id = o.customer_id  
LEFT JOIN order_items oi ON o.order_id = oi.order_id  
GROUP BY c.customer_id, c.first_name, c.last_name  
ORDER BY "Total Amount Spent" DESC;
```

customer_id	Customer Name	Total Amount Spent
1	Ali Khan	2175.0
4	Maria Garcia	150.0
3	John Smith	60.0
2	Sara Ahmed	40.0
5	Chen Wei	

Operation 06

```
SELECT c.customer_id,  
       c.first_name || ' ' || c.last_name AS "Customer Name",  
       COUNT(o.order_id) AS "Number of Orders"  
FROM customers c  
JOIN orders o ON c.customer_id = o.customer_id  
GROUP BY c.customer_id, c.first_name, c.last_name  
HAVING COUNT(o.order_id) > 1;
```

customer_id	Customer Name	Number of Orders
1	Ali Khan	2

Operation 07

```
SSELECT p.product_id,  
        p.product_name AS "Product Name",  
        SUM(oi.quantity * oi.unit_price) AS "Total Revenue"  
        FROM products p  
LEFT JOIN order_items oi ON p.product_id = oi.product_id  
        GROUP BY p.product_id, p.product_name  
        ORDER BY "Total Revenue" DESC;
```

product_id	Product Name	Total Revenue
1	Laptop Pro 15	1200.0
6	Smartphone X	900.0
4	Winter Jacket	150.0
2	Wireless Mouse	75.0
7	Novel: The Last Query	60.0
3	Database Design Book	40.0
5	Microwave Oven	

Operation 08

```
SELECT p.product_id,  
        p.product_name AS "Product Name",  
        p.price AS "Price"  
        FROM products p  
WHERE p.price = (SELECT MAX(price) FROM products);
```

product_id	Product Name	Price
1	Laptop Pro 15	1200.0

Operation 09

```
SELECT c.customer_id,  
       c.first_name || ' ' || c.last_name AS "Customer Name",  
       o.order_id AS "Order ID",  
       o.order_date AS "Order Date",  
       o.status AS "Order Status"  
FROM customers c  
LEFT JOIN orders o ON c.customer_id = o.customer_id;  
  
HAVING SUM(global_sales_million) > 100;
```

customer_id	Customer Name	Order ID	Order Date	Order Status
1	Ali Khan	1	2023-10-01	Shipped
1	Ali Khan	3	2023-11-10	Delivered
2	Sara Ahmed	2	2023-10-05	Pending
3	John Smith	4	2023-11-15	Cancelled
4	Maria Garcia	5	2023-12-01	Delivered
5	Chen Wei			

Operation 10

```
SELECT p.product_id,  
       p.product_name AS "Product Name",  
       CASE  
         WHEN oi.product_id IS NOT NULL THEN 'Yes'  
         ELSE 'No'  
       END AS "Ever Ordered",  
       COUNT(DISTINCT oi.order_id) AS "Times Ordered"  
FROM products p  
LEFT JOIN order_items oi ON p.product_id = oi.product_id  
GROUP BY p.product_id, p.product_name;
```

product_id	Product Name	Ever Ordered	Times Ordered
1	Laptop Pro 15	Yes	1
2	Wireless Mouse	Yes	2
3	Database Design Book	Yes	1
4	Winter Jacket	Yes	1
5	Microwave Oven	No	0
6	Smartphone X	Yes	1
7	Novel: The Last Query	Yes	1

Operation 11

```

SELECT s.supplier_id,
       s.supplier_name AS "Supplier Name",
       p.product_id,
       p.product_name AS "Product Name"
FROM suppliers s
LEFT JOIN products p ON s.supplier_id = p.supplier_id
ORDER BY s.supplier_id, p.product_id;

```

supplier_id	Supplier Name	product_id	Product Name
1	TechWorld Ltd	1	Laptop Pro 15
1	TechWorld Ltd	2	Wireless Mouse
1	TechWorld Ltd	6	Smartphone X
2	BookPlanet	3	Database Design Book
2	BookPlanet	7	Novel: The Last Query
3	FashionHub	4	Winter Jacket
4	HomePro	5	Microwave Oven

Operation 12

```
SELECT c.customer_id,  
       c.first_name || ' ' || c.last_name AS "Customer Name",  
       o.order_id AS "Order ID",  
       o.order_date,  
       o.status,  
       p.payment_id AS "Payment ID",  
       p.amount AS "Payment Amount",  
       CASE  
         WHEN o.order_id IS NOT NULL AND p.payment_id IS NULL THEN 'Unpaid  
           Order'  
         WHEN o.order_id IS NULL AND p.payment_id IS NOT NULL THEN 'Orphan  
           Payment'  
         WHEN o.order_id IS NOT NULL AND p.payment_id IS NOT NULL THEN 'Paid  
           Order'  
         ELSE 'No Order/No Payment'  
       END AS "Status"  
FROM customers c  
LEFT JOIN orders o ON c.customer_id = o.customer_id  
FULL JOIN payments p ON o.order_id = p.order_id;
```

customer_id	Customer Name	Order ID	order_date	status	Payment ID	Payment Amount	Status
1	Ali Khan	1	2023-10-01	Shipped	1	1250.0	Paid Order
1	Ali Khan	3	2023-11-10	Delivered	2	925.0	Paid Order
2	Sara Ahmed	2	2023-10-05	Pending			Unpaid Order
3	John Smith	4	2023-11-15	Cancelled			Unpaid Order
4	Maria Garcia	5	2023-12-01	Delivered	3	150.0	Paid Order
5	Chen Wei						No Order/No Payment

Operation 13

```
SELECT s1.supplier_name AS "Supplier 1",  
       s2.supplier_name AS "Supplier 2",  
       s1.country AS "Country",  
       p1.product_name AS "Product from Supplier 1",  
       p2.product_name AS "Product from Supplier 2"  
FROM suppliers s1  
JOIN suppliers s2 ON s1.country = s2.country AND s1.supplier_id < s2.supplier_id  
LEFT JOIN products p1 ON s1.supplier_id = p1.supplier_id  
LEFT JOIN products p2 ON s2.supplier_id = p2.supplier_id  
ORDER BY s1.country, s1.supplier_name;
```


Operation 14

```
SELECT c.customer_id,  
       c.first_name || ' ' || c.last_name AS "Customer Name",  
       SUM(oi.quantity * oi.unit_price) AS "Total Spending"  
FROM customers c  
JOIN orders o ON c.customer_id = o.customer_id  
JOIN order_items oi ON o.order_id = oi.order_id  
GROUP BY c.customer_id, c.first_name, c.last_name  
ORDER BY "Total Spending" DESC  
LIMIT 3;
```

customer_id	Customer Name	Total Spending
1	Ali Khan	2175.0
4	Maria Garcia	150.0
3	John Smith	60.0

Operation 15

```
SELECT  
CASE STRFTIME('%m', o.order_date)  
  WHEN '01' THEN 'January'  
  WHEN '02' THEN 'February'  
  WHEN '03' THEN 'March'  
  WHEN '04' THEN 'April'  
  WHEN '05' THEN 'May'  
  WHEN '06' THEN 'June'  
  WHEN '07' THEN 'July'  
  WHEN '08' THEN 'August'  
  WHEN '09' THEN 'September'  
  WHEN '10' THEN 'October'  
  WHEN '11' THEN 'November'  
  WHEN '12' THEN 'December'
```

```

END || ' ' || STRFTIME('%Y', o.order_date) AS "Month",
SUM(oi.quantity * oi.unit_price) AS "Total Sales Revenue"
FROM orders o
JOIN order_items oi ON o.order_id = oi.order_id
WHERE o.status != 'Cancelled'
GROUP BY STRFTIME('%Y-%m', o.order_date)
ORDER BY STRFTIME('%Y-%m', o.order_date);

```

Month	Total Sales Revenue
October 2023	1290.0
November 2023	925.0
December 2023	150.0